

WTW158 Standaard integrale
STANDAARD INTEGRALE

1. Integraal van 'n konstante funksie

$$\int k dx = kx + C, \quad k \in \mathbb{R}$$

2. Integraal van 'n magfunksie

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C$$

$$\int [f(x)]^n f'(x) dx = \frac{1}{n+1} [f(x)]^{n+1} + C$$

3. Integraal van die natuurlike eksponensiaalfunksie

$$\int e^x dx = e^x + C$$

$$\int e^{f(x)} f'(x) dx = e^{f(x)} + C$$

4. Integraal van 'n eksponensiaalfunksie

$$\int a^x dx = \frac{1}{\ln a} a^x + C, \quad a > 0$$

$$\int a^{f(x)} f'(x) dx = \frac{1}{\ln a} a^{f(x)} + C, \quad a > 0$$

5. Die integrale van die trigonometriese funksies

$$\int \sin x dx = -\cos x + C$$

$$\int \sin(f(x)) f'(x) dx = -\cos(f(x)) + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \cos(f(x)) f'(x) dx = \sin(f(x)) + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \sec^2(f(x)) f'(x) dx = \tan(f(x)) + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \operatorname{cosec}(f(x)) \cot(f(x)) f'(x) dx = -\operatorname{cosec}(f(x)) + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \sec(f(x)) \tan(f(x)) f'(x) dx = \sec(f(x)) + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \operatorname{cosec}^2(f(x)) f'(x) dx = -\cot(f(x)) + C$$

6. Die integrale van die inverse trigonometriese funksies

$$\int \frac{1}{\sqrt{1-x^2}} dx = \text{bgsin}(x) + C = \sin^{-1}(x) + C$$

$$\int \frac{f'(x)}{\sqrt{1-[f(x)]^2}} dx = \text{bgsin}(f(x)) + C = \sin^{-1}(f(x)) + C$$

$$\int \frac{1}{1+x^2} dx = \text{bgtan}(x) + C = \tan^{-1}(x) + C$$

$$\int \frac{f'(x)}{1+[f(x)]^2} dx = \text{bgtan}(f(x)) + C = \tan^{-1}(f(x)) + C$$

7. Die integrale van logaritmiese funksies

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

8. Die integrale van hiperboliese funksies

$$\int \sinh x dx = \cosh x + C$$

$$\int \sinh f(x) f'(x) dx = \cosh f(x) + C$$

$$\int \cosh x dx = \sinh x + C$$

$$\int \cosh f(x) f'(x) dx = \sinh f(x) + C$$